

APR8025 Data Analytics

Fall 2026

Instructors: Hye Hyun Ma, gogus8281@hanyang.ac.kr

Class: Thu 14:00 - 17:00 in Y202-502 Lecture Room

Office hours: Thu 9:00 - 12:00 (Any time by appointment)

Course Description

This course is designed for graduate students who conduct quantitative research in the social sciences. You will learn the basic concepts and methods of data analytics needed to conduct quantitative research, with an emphasis on choosing appropriate analytical methods for your research questions and interpreting and reporting analytical results in a transparent manner. This semester will focus on basic statistical analysis and issues including descriptive statistics and inferential statistics such as t-test, chi-squared test, analysis of variance, regression. In all assignments and exams there is an emphasis on development of written expository skills as an integral part of research analysis.

Course Objectives

In this course, you will learn how data analysis is conducted in quantitative research and how to choose analytical methods that fit your research questions and data characteristics. You will practice conducting basic statistical analyses using R, interpreting analytical results, and reporting the results in a manner appropriate for scholarly journal articles. The course combines lectures, weekly problem sets, and brief research article discussions.

Course Material

- 이병관 (2023). 사회과학 연구자를 위한 R을 활용한 기초통계. 한양대학교 출판부

Grades

1. Weekly problem sets (40%)
2. Final exam (50%)
3. Attendance (10%)

Schedule of Topics and Readings

W01 (3 Sep) Orientation

- ❖ Boster, F. J. (2002). On making progress in communication science. *Human Communication Research*, 28(4), 473–490.
- ❖ 이병관, 김재민, 김주환, 장다연, & 권나현. (2020). 국내 커뮤니케이션 연구의 통계분석방법의 현황과 문제점: 지난 10년간 「한국언론학보」 게재 논문의 내용분석. *한국언론학보*, 64(5), 118–152.

W02 (10 Sep) Introduction to R and R Programming Basic

W03 (17 Sep) Descriptive Statistics — Problem Set #1

- ❖ Westfall, P. H. (2014). Kurtosis as peakedness, 1905–2014. RIP. *The American Statistician*, 68(3), 191–195.
- ❖ Gorard, S. (2005). Revisiting a 90-year-old debate: The advantages of the mean deviation. *British Journal of Educational Studies*, 53(4), 417–430.

W04 (24 Sep) Thanksgiving Holiday

W05 (1 Oct) Inferential Statistics I: Sampling distributions and hypothesis testing — Problem Set #2

- ❖ Hunter, J. E. (1997). Needed: A ban on the significance test. *Psychological Science*, 8, 3–7.
- ❖ Cohen, J. (1994). The earth is round ($p < .05$). *American Psychologist*, 49(12), 997–1003.
- ❖ Schmidt, F. L., & Hunter, J. E. (1997). Eight common but false objections to the discontinuation of significance testing in the analysis of research data. In L. L. Harlow, S. A. Mulaik, & J. H. Steiger (Eds.), *What if there were no significance tests?* (pp. 37–64). Lawrence Erlbaum Associates.

W06 (8 Oct) Research Design, Statistical Planning, and Open Science

- ❖ Wicherts, J. M., Veldkamp, C. L., Augusteyn, H. E., Bakker, M., Van Aert, R. C., & Van Assen, M. A. (2016). Degrees of freedom in planning, running, analyzing, and reporting psychological studies: A checklist to avoid p-hacking. *Frontiers in psychology*, 7, 1832.
- ❖ Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological science*, 22(11), 1359–1366.

W07 (15 Oct) Inferential Statistics II: t-test— *Problem Set #3*

- ❖ 이병관, 마혜현, 김엘레나, 방서여, 김태수. (2025). 국내 커뮤니케이션 연구에서의 연구 재현성: 출판 편향과 p-해킹의 실증 연구. *한국언론학보*, 69(4), 36-71.
- ❖ O'Brien, S. F., & Yi, Q. L. (2016). How do I interpret a confidence interval? *Transfusion*, 56(7), 1680–1683.
- ❖ Kirk, R. E. (1996). Practical significance: A concept whose time has come. *Educational and Psychological Measurement*, 56(5), 746–759.

W08 (22 Oct) Chi-Squared Test — *Problem Set #4*

- ❖ Lombard, M., Snyder-Duch, J., & Bracken, C. C. (2002). Content analysis in mass communication: Assessment and reporting of intercoder reliability. *Human Communication Research*, 28(4), 587–604.
- ❖ Kroonenberg, P. M., & Verbeek, A. (2018). The tale of Cochran's rule: My contingency table has so many expected values smaller than 5, what am I to do? *The American Statistician*, 72(2), 175–183.

W09 (29 Oct) ANOVA I — *Problem Set #5*

- ❖ Levine, T. R., & Hullett, C. R. (2002). Eta squared, partial eta squared, and misreporting of effect size in communication research. *Human Communication Research*, 28(4), 612–625.
- ❖ Rossi, J. S. (1990). Statistical power of psychological research: What have we gained in 20 years? *Journal of Consulting and Clinical Psychology*, 58(5), 646–656.

W10 (5 Nov) ANOVA II — *Problem Set #6*

- ❖ Cornell, J. E., Young, D. M., Seaman, S. L., & Kirk, R. E. (1992). Power comparisons of eight tests for sphericity in repeated measures designs. *Journal of Educational Statistics*, 17(3), 233–249.
- ❖ Muller, K. E., & Barton, C. N. (1989). Approximate power for repeated-measures ANOVA lacking sphericity. *Journal of the American Statistical Association*, 84(406), 549–555.
- ❖ Tybout, A., & Sternthal, B. (2001). Can I test for simple effects in the presence of an insignificant interaction? *Journal of Consumer Psychology*, 5–10.
- ❖ Rosnow, R. L., & Rosenthal, R. (1989). Definition and interpretation of interaction effects. *Psychological Bulletin*, 105(1), 143–146.
- ❖ Meyer, D. L. (1991). Misinterpretation of interaction effects: A reply to Rosnow and Rosenthal. *Psychological Bulletin*, 110(3), 571–573.

W11 (12 Nov) Correlation and Simple Regression — *Problem Set #7*

- ❖ Hunter, J. E., & Schmidt, F. L. (2004). *Methods of meta-analysis: Correcting error and bias in research findings*. Sage.

- ❖ Sudman, S., et al. (2001). Other statistical issues. *Journal of Consumer Psychology*, 10(1–2), 110–112.

W12 (19 Nov) Multiple Regression I — *Problem Set #8*

- ❖ Mason, C. H., & Perreault, W. D., Jr. (1991). Collinearity, power, and interpretation of multiple regression analysis. *Journal of Marketing Research*, 28, 268–280.
- ❖ Jang, D., & Lee, B. (2022). When machine learning meets social science: A comparative study of ordinary least square, stochastic gradient descent, and support vector regression for exploring the determinants of behavioral intentions to tuberculosis screening. *Asian Communication Research*, 19(3), 101–118.

W13 (26 Nov) Multiple Regression II — *Problem Set #9*

- ❖ Echambadi, R., & Hess, J. D. (2007). Mean-centering does not alleviate collinearity problems in moderated multiple regression models. *Marketing Science*, 26(3), 438–445.

W14 (3 Dec) Regression Diagnostics — *Problem Set #10*

- ❖ Chatterjee, S., & Yilmaz, M. (1992). A review of regression diagnostics for behavioral research. *Applied Psychological Measurement*, 16(3), 209–227.
- ❖ Farrar, D. E., & Glauber, R. R. (1967). Multicollinearity in regression analysis: The problem revisited. *The Review of Economics and Statistics*, 92–107.
- ❖ Williams, M. N., Grajales, C. A. G., & Kurkiewicz, D. (2013). Assumptions of multiple regression: Correcting two misconceptions. *Practical Assessment, Research, and Evaluation*, 18(1), Article 11.

W15 (10 Dec) Final Exam (In-class)

W16 (17 Dec) Final Wrap-up

The above schedule may be subject to change according to the status of the class.

Course Policies

1. AI Policy

You are welcome to use generative AI tools in this class. However, any use of AI in submitted work must be clearly acknowledged and properly documented. You are responsible for verifying the accuracy, originality, and appropriate use of AI-generated content.

For each assignment, please include a brief “AI Use Disclosure” indicating:

- Generative AI tool and model used:
 - ChatGPT (GPT-5)
 - Claude (Claude Sonnet 4)
 - Gemini (Gemini 2.5 Pro)
 - Other: _____
- Purpose(s) of use:
 - Idea generation and exploration
 - Copyediting, language improvement, or translation
 - LLM-assisted information search
 - Literature search or classification
 - Coding assistance or debugging
 - Data analysis assistance
 - Other: _____

2. Academic Integrity Policy

I have a responsibility to uphold the university’s academic integrity policy, and I take that responsibility very seriously. If you are unsure whether a particular form of assistance is acceptable, please ask me before submitting your work.

Discussion of concepts with classmates and the use of external resources or AI tools for learning are permitted. However, copying or submitting work produced by another person, source, or AI as your own is not permitted. This course maintains a **zero-tolerance policy toward cheating and plagiarism. If you cannot explain your answer, reproduce the reasoning, or complete a similar task independently, the submitted work will receive a grade of zero.** After submission, claiming that you “did not know” or were unaware of the policy will not exempt you from responsibility for an academic integrity violation.